Assignment 2

Name: _____ Student ID: _____ Marks:_____

We express $u(t, x)$ in terms of a new function $v(t, x)$ to solve the following boundary value problem:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} - 6\frac{\partial u}{\partial x}, \quad 0 \leq x \leq \pi, \quad t \geq 0,$$
$$u(t, 0) = 0, \quad u(t, \pi) = 0, \quad u(0, x) = x(\pi - x).$$

(a) First, state the new boundary value problem in terms of $v(t, x)$, including the coordinate ranges and boundary conditions. [10]

(b) Then, find the general solution consistent with the first two boundary conditions. (You do not need to use the third condition to fix the final solution). [10]